

Notes: ‘n’ represents the dimension of the corresponding problem. ‘NI’ represents the total number of iterations. ‘NF’ represents the number of the function evaluation. ‘T’ represents the running time of the algorithm. ‘ $\|g^*\|$ ’ represents the gradient value for the solution. The values of ‘NI’, ‘NF’, ‘T’ and ‘ $\|g^*\|$ ’ are set as ‘NaN’ if and only if the number of iterations is greater than 2000.

Table 1: Numerical results for Test 1				
Problem/n	SHCG-V1 NI/NF/T/ $\ g^*\ $	SHCG-V2 NI/NF/T/ $\ g^*\ $	SHCG-V3 NI/NF/T/ $\ g^*\ $	SHCG-V4 NI/NF/T/ $\ g^*\ $
dixmaana/3000	19/83/0.061/1.96e-07	24/88/0.064/4.84e-07	21/81/0.065/9.15e-07	22/83/0.061/6.63e-07
dixmaana/30000	17/86/0.507/3.56e-07	18/87/0.515/4.67e-07	20/86/0.511/9.75e-07	25/94/0.559/7.64e-07
dixmaana/600000	22/97/11.009/9.49e-07	25/98/11.303/4.44e-07	14/84/9.811/6.03e-07	20/96/11.214/4.88e-07
dixmaanb/6000	21/88/0.118/2.93e-07	18/85/0.117/1.28e-07	18/84/0.110/6.07e-07	16/80/0.108/7.42e-07
dixmaanb/15000	26/93/0.303/5.66e-07	24/91/0.289/6.90e-07	20/87/0.274/8.58e-08	23/91/0.295/9.48e-08
dixmaanb/60000	19/89/0.973/9.11e-07	22/92/1.015/1.22e-07	23/93/1.047/6.05e-07	17/83/0.892/7.77e-07
dixmaanc/3000	21/80/0.054/8.82e-07	17/82/0.054/1.09e-07	18/81/0.059/9.54e-07	14/79/0.052/8.38e-07
dixmaanc/150000	23/97/2.867/6.40e-07	20/93/2.777/7.69e-07	21/92/2.715/9.06e-07	24/97/2.879/4.42e-07
dixmaand/300000	15/86/4.970/3.48e-07	19/94/5.460/3.16e-07	19/91/5.304/6.66e-07	29/104/6.058/8.72e-07
dixmaane/1500	297/413/0.169/7.81e-07	363/426/0.161/9.23e-07	315/399/0.160/6.24e-07	357/419/0.164/7.91e-07
dixmaane/3000	431/601/0.409/9.98e-07	536/602/0.408/4.04e-07	478/583/0.411/8.63e-07	477/541/0.373/9.94e-07
dixmaane/15000	820/1054/3.487/8.37e-07	1115/1188/3.872/9.82e-07	825/1039/3.408/7.91e-07	997/1064/3.459/9.81e-07
dixmaanf/1500	270/391/0.164/9.23e-07	312/388/0.131/9.22e-07	265/358/0.129/8.35e-07	363/427/0.155/9.80e-07
dixmaanf/3000	290/409/0.270/7.05e-07	390/459/0.307/9.49e-07	359/466/0.319/7.27e-07	364/428/0.290/9.27e-07
dixmaanf/30000	838/1147/6.499/8.98e-07	1114/1200/6.909/9.18e-07	783/996/5.655/6.94e-07	1627/1723/9.815/7.51e-07
dixmaang/300	128/203/0.020/8.32e-07	202/274/0.026/8.99e-07	158/233/0.022/8.75e-07	279/338/0.033/9.76e-07
dixmaang/1500	250/357/0.119/9.69e-07	341/422/0.139/9.63e-07	332/432/0.146/8.96e-07	502/571/0.188/9.00e-07
dixmaang/15000	1609/2151/6.782/4.16e-07	1825/2118/6.731/9.81e-07	1648/1986/6.244/8.60e-07	NaN/NaN/NaN/NaN
dixmaanh/15000	798/1089/3.427/6.72e-07	1130/1233/3.877/9.94e-07	719/889/2.801/9.80e-07	932/1017/3.239/9.02e-07
dixmaanh/21000	613/885/3.689/7.40e-07	902/991/4.187/9.59e-07	1984/4380/18.379/8.67e-07	1130/1234/5.235/9.71e-07
dixmaanh/30000	874/1178/6.710/8.42e-07	1224/1335/10.819/9.88e-07	1095/1358/7.746/8.46e-07	1157/1249/7.129/9.56e-07
dixmaani/30	217/319/0.024/8.31e-07	216/296/0.020/7.96e-07	203/266/0.046/9.81e-07	257/324/0.031/9.08e-07
dixmaani/90	595/797/0.071/9.48e-07	744/842/0.113/9.04e-07	632/746/0.066/8.92e-07	812/888/0.083/9.46e-07
dixmaani/210	1516/2041/0.325/7.77e-07	1341/1433/0.208/9.06e-07	1242/1452/0.316/8.82e-07	1224/1284/0.256/9.70e-07
dixmaanaj/600	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanaj/6000	858/1161/4.803/7.17e-07	NaN/NaN/NaN/NaN	745/925/3.525/6.73e-07	NaN/NaN/NaN/NaN
dixmaanl/300	1113/1523/0.138/8.07e-07	1812/1983/0.163/9.85e-07	1305/1572/0.131/7.68e-07	NaN/NaN/NaN/NaN
dixmaanl/21000	910/1234/5.017/6.05e-07	1879/1980/8.122/9.63e-07	952/1181/4.744/9.08e-07	NaN/NaN/NaN/NaN
dixmaanl/30000	831/1157/6.390/9.53e-07	1933/2020/11.227/9.80e-07	1141/1394/7.724/9.84e-07	1702/1775/9.889/9.78e-07
dixon3dq/50	524/735/0.009/5.65e-07	498/600/0.007/6.78e-07	488/623/0.007/9.40e-07	592/670/0.008/8.22e-07
dixon3dq/100	936/1260/0.013/9.01e-07	1055/1186/0.012/8.49e-07	936/1144/0.012/8.27e-07	779/841/0.009/8.10e-07
dixon3dq/150	1785/2406/0.022/9.68e-07	1725/1877/0.017/9.73e-07	1815/2184/0.022/8.29e-07	1378/1432/0.014/8.41e-07
dqdrtic/10000	95/226/0.022/6.95e-07	118/339/0.027/5.47e-07	77/228/0.016/5.19e-07	111/262/0.020/5.89e-07
dqdrtic/50000	80/206/0.071/7.68e-07	100/297/0.102/7.28e-07	100/294/0.101/7.26e-07	119/293/0.099/9.05e-07
dqdrtic/150000	95/232/0.295/7.57e-07	103/295/0.356/7.38e-07	76/221/0.269/3.47e-07	117/312/0.380/6.64e-07
dqrtic/5000	54/164/0.165/3.14e-07	55/162/0.166/8.83e-07	56/168/0.170/6.27e-07	61/173/0.176/6.38e-07
edensch/500	30/105/0.012/2.61e-07	49/123/0.013/5.82e-07	44/195/0.020/8.63e-07	53/150/0.015/7.67e-07
edensch/20000	67/385/1.525/7.93e-07	62/267/1.060/8.16e-07	78/502/1.979/4.01e-07	66/229/0.908/8.00e-07
edensch/120000	210/1663/82.237/8.37e-07	127/853/43.054/9.84e-07	69/375/21.556/6.99e-07	121/751/42.644/5.46e-07
eg2/100	236/752/0.032/8.64e-07	345/1456/0.071/8.82e-07	303/1422/0.060/7.68e-07	171/560/0.023/5.65e-07
eg2/200	NaN/NaN/NaN/NaN	1399/12379/1.075/7.97e-07	343/1322/0.225/8.24e-07	431/2338/0.261/8.53e-07
fletcher/500	99/703/0.036/9.63e-07	281/2425/0.101/2.83e-07	115/887/0.040/7.33e-07	254/1904/0.087/3.46e-07
fletcher/7000	139/1147/0.148/4.75e-07	295/2731/0.327/4.77e-07	260/2188/0.271/7.02e-07	446/4053/0.525/1.11e-07
freuroth/50	218/1174/0.043/9.11e-07	218/869/0.029/6.00e-07	181/550/0.021/8.05e-07	404/1985/0.068/7.64e-07
freuroth/500	NaN/NaN/NaN/NaN	1054/10047/0.611/3.89e-07	1087/10723/0.688/9.94e-07	903/8118/0.361/9.75e-07
freuroth/5000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	923/9512/1.707/7.43e-07	NaN/NaN/NaN/NaN
genrose/10000	234/392/0.075/5.52e-07	311/495/0.098/9.75e-07	290/462/0.097/9.99e-07	380/519/0.107/9.64e-07
himmelbg/1000	2/13/0.002/7.55e-30	2/13/0.001/7.54e-30	2/13/0.001/7.55e-30	2/13/0.001/7.55e-30
himmelbg/70000	2/15/0.108/8.37e-43	2/15/0.106/8.34e-43	2/15/0.086/8.37e-43	2/15/0.119/8.37e-43
himmelbg/600000	3/16/0.720/5.32e-26	3/16/0.731/1.39e-26	3/16/0.768/1.30e-26	3/16/0.700/1.61e-26
liarwhd/10000	233/680/0.097/3.52e-07	133/488/0.058/9.39e-07	128/456/0.104/3.94e-07	124/455/0.061/8.36e-07
liarwhd/30000	143/537/0.250/7.87e-07	141/607/0.270/1.95e-07	164/562/0.250/5.06e-07	136/577/0.255/9.63e-07
liarwhd/120000	305/882/1.040/3.70e-07	155/683/0.736/6.83e-07	164/651/0.699/1.51e-07	133/594/0.637/5.64e-07
penalty1/100	18/108/0.007/6.16e-07	19/109/0.006/8.92e-07	19/109/0.006/8.13e-07	19/109/0.006/9.29e-07
penalty1/10000	20/94/11.483/2.94e-09	20/92/11.841/2.06e-07	20/92/10.400/4.52e-07	20/92/11.066/1.83e-07
quartc/5000	54/164/0.263/3.14e-07	55/162/0.262/8.83e-07	56/168/0.260/6.27e-07	61/173/0.268/6.38e-07
tridia/100	361/489/0.023/9.50e-07	349/429/0.017/9.49e-07	251/341/0.013/8.82e-07	295/364/0.018/9.16e-07
tridia/300	591/793/0.032/8.11e-07	735/819/0.030/9.61e-07	623/768/0.035/9.58e-07	512/585/0.031/9.22e-07
tridia/1500	1675/2260/0.176/9.30e-07	1763/1855/0.187/9.51e-07	1960/2355/0.201/9.73e-07	1482/1567/0.099/9.96e-07
woods/5000	196/534/0.052/7.03e-07	219/499/0.045/6.96e-07	158/434/0.046/8.84e-07	300/578/0.082/4.16e-07
woods/300000	230/578/4.991/2.66e-07	252/656/5.661/9.23e-07	210/530/4.204/8.38e-07	300/650/5.710/2.03e-07
woods/1000000	179/436/10.749/4.15e-07	391/826/17.086/5.49e-07	218/648/15.261/5.72e-07	376/750/19.130/6.87e-07
bdexp/1000	2/11/0.003/2.98e-133	2/11/0.002/2.96e-133	2/11/0.003/2.99e-133	2/11/0.002/2.99e-133
bdexp/10000	2/17/0.017/2.62e-61	2/17/0.016/2.62e-61	2/17/0.016/2.62e-61	2/17/0.016/2.62e-61
bdexp/100000	2/12/0.297/4.57e-106	2/12/0.316/4.57e-106	2/12/0.295/4.57e-106	2/12/0.216/4.57e-106
exdenschnf/10000	38/114/0.020/5.03e-07	52/130/0.036/9.91e-07	40/113/0.019/8.83e-07	48/124/0.021/4.67e-07
exdenschnb/3000	25/89/0.010/5.69e-07	34/98/0.007/4.71e-07	26/89/0.011/9.86e-07	47/111/0.010/1.10e-07
exdenschnb/5000000	40/115/10.797/7.22e-07	26/108/9.655/1.91e-07	49/125/11.163/3.21e-07	36/111/9.680/4.17e-07
genquartic/50000	18/87/0.062/5.78e-07	28/97/0.070/6.25e-07	25/93/0.066/3.19e-07	29/97/0.071/8.47e-07
genquartic/100000	19/81/0.088/5.37e-07	28/98/0.107/6.85e-07	21/86/0.102/7.90e-07	24/93/0.100/9.72e-07
genquartic/200000	0/0/0.003/0.00e+00	0/0/0.002/0.00e+00	0/0/0.002/0.00e+00	0/0/0.003/0.00e+00
biggsb1/250	1282/1688/0.077/7.82e-07	1613/1778/0.083/9.16e-07	1518/1838/0.068/9.91e-07	1156/1214/0.057/7.95e-07
fletcbv3/100	873/1724/0.150/9.53e-07	1681/3820/0.313/7.28e-07	986/2159/0.170/7.87e-07	NaN/NaN/NaN/NaN
nonscomp/10000	NaN/NaN/NaN/NaN	351/679/0.116/7.60e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
nonscomp/70000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1774/2707/2.690/8.13e-07	NaN/NaN/NaN/NaN
nonscomp/100000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1831/2767/3.728/8.60e-07	NaN/NaN/NaN/NaN
power1/100	1137/1545/0.136/7.45e-07	1431/1582/0.092/9.79e-07	1360/1617/0.067/9.75e-07	1489/1589/0.070/8.07e-07
raydan1/300	137/205/0.012/5.48e-07	273/367/0.017/9.66e-07	173/249/0.018/9.08e-07	356/463/0.022/8.59e-07
raydan2/2000	14/71/0.012/7.00e-07	14/67/0.015/5.34e-07	14/71/0.010/7.00e-07	14/71/0.009/7.00e-07
raydan2/70000	21/102/1.072/2.05e-07	35/144/1.486/2.46e-07	12/72/0.745/2.93e-07	12/72/0.758/2.93e-07
diagonal1/250	1066/8565/0.767/8.36e-07	419/2919/0.230/7.89e-07	550/4644/0.294/8.94e-07	474/3601/0.279/9.33e-07
diagonal2/500	156/268/0.019/4.95e-07	217/329/0.022/9.36e-07	158/264/0.023/9.84e-07	253/382/0.027/6.22e-07
diagonal2/5000	463/741/0.317/9.56e-07	626/956/0.424/8.20e-07	535/842/0.387/7.01e-07	601/866/0.397/9.78e-07
diagonal2/10000	764/1191/0.966/9.71e-07	902/1328/1.078/6.71e-07	696/1043/0.849/5.81e-07	837/1195/0.971/9.18e-07
singx/300	307/812/0.193/5.69e-07	154/413/0.074/7.52e-07	161/438/0.102/9.85e-07	101/278/0.070/7.40e-07
singx/1000	225/528/1.951/4.63e-07	166/439/1.757/2.84e-07	243/620/1.960/7.79e-07	156/435/1.445/4.67e-07
singx/3000	188/460/13.004/7.87e-07	98/260/7.146/8.75e-07	347/769/23.321/4.80e-07	148/405/11.431/5.05e-07
lin/30	17/79/0.015/6.58e-07	17/79/0.018/7.20e-07	17/79/0.033/6.96e-07	17/79/0.013/7.02e-07
lin/400	14/72/0.300/7.97e-07	14/72/0.302/7.96e-07	14/72/0.216/7.97e-07	14/72/0.190/7.97e-07
lin/700	18/81/12.866/2.20e-07	18/81/11.692/2.85e-07	18/81/11.825/2.85e-07	18/81/13.156/2.85e-07
osb2/11	606/1011/0.075/7.30e-07	390/634/0.050/7.23e-07	414/708/0.063/8.22e-07	495/751/0.076/8.82e-07
pen1/50	136/510/0.028/7.03e-07	113/436/0.025/4.22e-07	67/315/0.016/8.38e-07	76/386/0.020/7.72e-07
pen1/100	172/555/0.054/7.82e-07	113/465/0.026/3.78e-07	60/329/0.024/8.95e-07	78/483/0.046/4.23e-07
pen1/600	161/1025/7.400/9.99e-07	91/576/3.231/8.56e-07	117/478/2.639/9.98e-07	95/484/3.534/9.95e-07
pen2/10	1245/3208/0.129/7.65e-07	755/2213/0.087/9.50e-07	846/2413/0.110/7.92e-07	519/1687/0.082/9.01e-07
pen2/50	381/1191/0.066/9.45e-07	304/1032/0.059/9.46e-07	287/921/0.055/6.02e-07	351/1235/0.068/8.77e-07
pen2/90	227/755/0.067/8.46e-07	183/506/0.048/6.33e-07	207/684/0.061/7.77e-07	220/739/0.071/3.70e-07
rosex/300	104/296/0.055/9.51e-07	80/305/0.041/3.26e-07	107/292/0.047/9.86e-07	116/393/0.084/7.55e-07
rosex/1000	99/348/1.904/9.04e-08	87/355/2.456/7.78e-07	107/340/2.008/9.72e-07	74/309/2.267/7.73e-07
trid/400	86/148/0.096/7.24e-07	109/170/0.101/9.73e-07	85/145/0.087/6.10e-07	134/194/0.106/9.32e-07